

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 1

VRAESTEL

Kopiereg voorbehou

Module EBN122

Elektrisiteit en Elektronika

August 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Department of Electrical, Electronic and
Computer Engineering

Semester Test 1

QUESTION PAPER

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Module EBN122

Electricity and Electronics

August 2017

Eksamensinligting:

Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		17 + OCR	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 at the start of the session.

Do step 1 in the first 90 minutes of the test

Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN122S1**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2						k	Ω								
04	-	1																	

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie.

Doen stap 1 in die eerste 90 minute van die toets.

Doen stap 2 binne die daaropvolgende 5 minute.

STAP 0 (-5 min to 0 min): Voltooi die voorblad van di OKH vorm

STAP 1 (0 min to 90 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (95 min to 100 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die "Assessment ID" is **2017EBN122S1**
- Sluit die korrekte eenheid vir alle antwoorde in. Maak gebruik van die toepaslike voorvoegsel.
- Gebruik 'n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

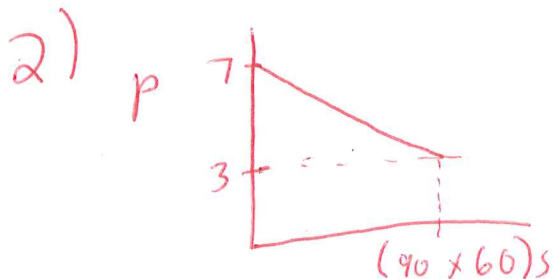
Voorbeelde

01	-	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2						k	Ω								
04	-	1																	

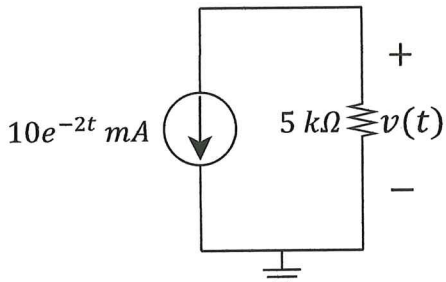
SECTION / AFDELING 1 [8]

Question 1 & 2 [4]		Vraag 1 & 2 [4]	
01	A house has twenty old 100 W tungsten lightbulbs which are switched on for 12 hours per day. How much money can the owners of the house save per year (365 days) if they replace the globes with new 20 W lightbulbs? Electricity costs R5/kWh.	01	'n Huis het twintig ou 100 W tungsten gloeilampe wat elk vir 12 ure per dag aan is. Hoeveel geld kan die eienaar van die huis in 'n jaar (365 dae) spaar indien hy die gloeilampe met nuwe 20 W gloeilampe vervang? Elektrisiteit kos R5/kWh.
02	A 12 V battery is used to power a remote controlled car. The power delivered by the battery falls linearly from 7 W to 3 W in 90 minutes. Determine the total energy usage of the battery over 90 minutes.	02	'n 12 V battery dryf 'n afstandsbeheerde karretjie aan. Die drywing wat die battery lewer val linêr vanaf 7 W tot 3 W in 90 minute. Wat is die totale energy verbruik van die battery oor 90 minute?

$$1) \text{ Saving} = R5/kWh \times \frac{(100-20)}{1000} kW \times 12 h \times 365 d \times 20 \\ = R 35040$$



$$\therefore W = \frac{1}{2}(7-3)(90)(60) + (3)(90)(60) \\ = 27 kJ$$

Question 3 & 4 [4]		Vraag 3 & 4 [4]	
03	The charge entering the positive terminal of an element is $q(t) = 10 \sin(4\pi t)$ mC. The voltage across the element is $v(t) = 2 \cos(4\pi t)$ V. Find the power delivered to the element at $t = 0.3$ s.	03	Die lading wat by die positiewe terminaal van 'n element ingaan is $q(t) = 10 \sin(4\pi t)$ mC. Die spanning oor die element is $v(t) = 2 \cos(4\pi t)$ V
04	For the circuit below, calculate the energy delivered to the resistor between 0 and 0.5 s.	04	Vir onderstaande kringbaan, bereken die energie gelewer aan die resistor tussen 0 en 0.5 s.
Circuit for Q04 / Kringbaan vir Q04 			

$$3) i = \frac{dq}{dt} = \frac{d}{dt} (10 \sin(4\pi t) \text{ mC})$$

$$= 40\pi \cos(4\pi t) \text{ mA}$$

$$p = vi = 40\pi \cos(4\pi t) \times 10^{-3} \times 2 \cos(4\pi t)$$

$$= 80\pi \cos^2(4\pi t) \text{ mW}$$

$$p(0,3) = 164,5 \text{ mW}$$

$$4) v(t) = -iR = -(10e^{-2t} \text{ mA})(5 \text{ k})$$

$$= -50e^{-2t} \text{ V}$$

$$W = \int_0^{0,5} p dt = \int_0^{0,5} -(10e^{-2t} \text{ mA})(-50e^{-2t}) dt$$

$$= 500 \text{ m} \int_0^{0,5} e^{-4t} dt = \frac{500 \text{ m}}{-4} [e^{-4t}]_0^{0,5}$$

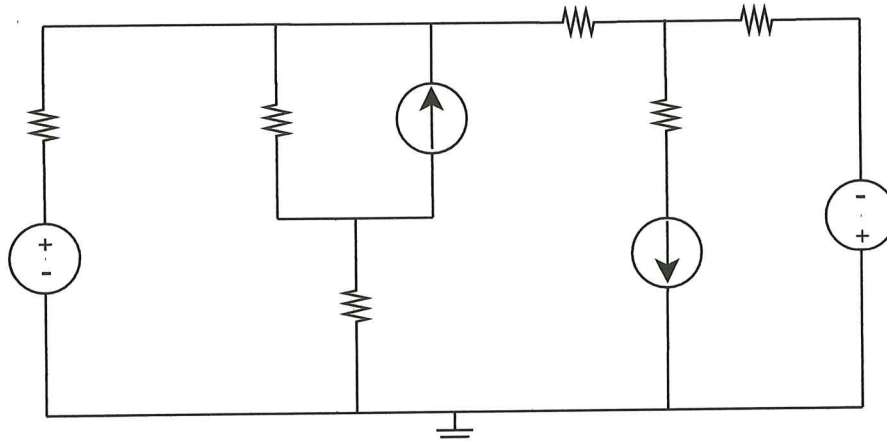
$$= -125 \text{ m} [e^{-4(0,5)} - e^0]$$

$$= 108,08 \text{ mJ}$$

SECTION / AFDELING 2 [14]

Question 5 & 6 [2]		Vraag 5 & 6 [2]	
05	How many <i>independent loops</i> does a circuit with 5 nodes and 9 branches have?	05	Hoeveel <i>onafhanklike lusse</i> het 'n kringbaan wat 5 nodusse en 9 takke het?
06	How many nodes does the circuit below have?	06	Hoeveel nodusse het onderstaande kringbaan?

Circuit for Q06 / Kringbaan vir Q06.



$$5) b = l + n - 1$$

$$l = 9 - 5 + 1 = 5$$

6) 7 nodes

Question 7 & 8 [4]		Vraag 7 & 8 [4]	
07	Calculate v_1 .	07	Bereken v_1 .
08	Find the power associated with component d .	08	Bepaal die drywing geassocieer met komponent d .

$$V_3 = -7V; \quad V_2 = 3V$$

$$-V_1 + V_2 - V_3 + 5 = 0$$

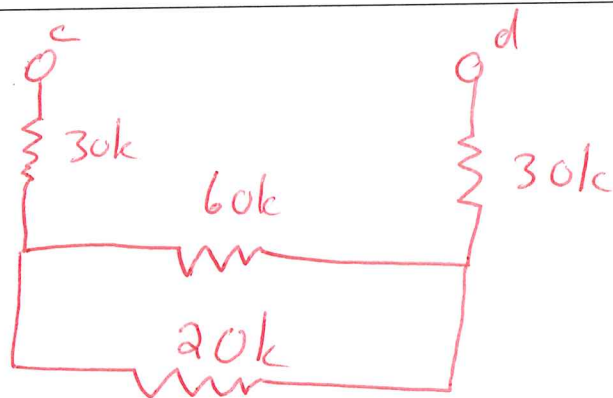
$$\therefore V_1 = 3 + 7 + 5 = 15V$$

$$i_1 + 1 = 2$$

$$i_1 = 1A$$

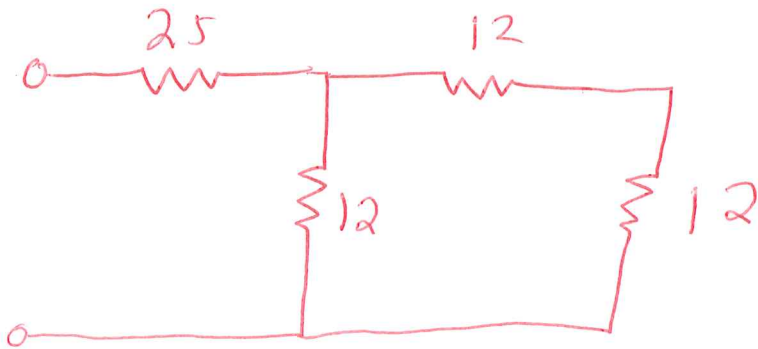
$$P_d = + (5)(1) = 5W$$

Question 9 [2]		Vraag 9 [2]	
09	Calculate the equivalent resistance with respect to terminals c-d.	09	Bereken die ekwivalente weerstande met verwysing tot terminale c-d.



$$\begin{aligned}
 \therefore R_{cd} &= 20 // 60 + 30 + 30 \\
 &= 75 \text{ k}\Omega
 \end{aligned}$$

Question 10 [2]		Vraag 10 [2]	
10	Calculate R_{ef} .	11	Bepaal R_{ef} .

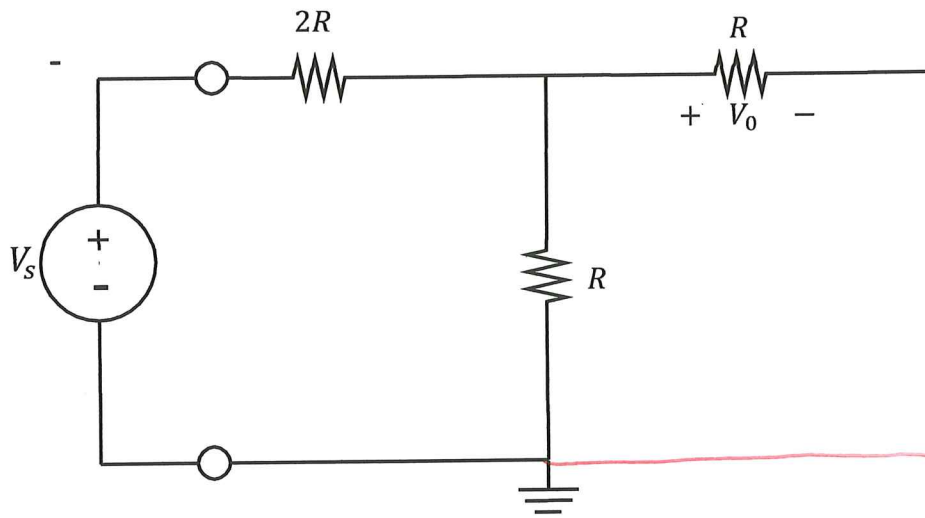


$$\therefore R_{ef} = 25 + 12 // 24$$

$$= 33 \Omega$$

Question 11 [2]

Vraag 11 [2]

11 Find V_0/V_s .11 Vind V_0/V_s .

$$V_0 = \frac{\frac{R}{2}}{2R + \frac{R}{2}} V_s$$

$$\begin{aligned} \frac{V_0}{V_s} &= \frac{R}{4R + R} \\ &= \frac{R}{5R} \\ &= \frac{1}{5} \end{aligned}$$

$$\frac{V_0}{V_s} = 0,2$$

Question 12 [2]		Vraag 12 [2]	
12	Calculate I_S if $V_1 = 3$ V.	12	Bereken I_S indien $V_1 = 3$ V.

$$I_2 = \frac{3}{20} = 0,15$$

$$I_2 = \frac{30}{110} (-I_S)$$

$$I_S = -\frac{110}{30} (0,15)$$

$$= -0,55 \text{ A}$$

SECTION / AFDELING 3 [24]

Question 13 & 14 [4]		Vraag 13 & 14 [4]	
13	Determine b . $\begin{bmatrix} 8 & -4 \\ -4 & a \end{bmatrix} \begin{bmatrix} 0.5b \\ 3 \end{bmatrix} = \begin{bmatrix} 2a \\ b \end{bmatrix}$	13	Bepaal b . $\begin{bmatrix} 8 & -4 \\ -4 & a \end{bmatrix} \begin{bmatrix} 0.5b \\ 3 \end{bmatrix} = \begin{bmatrix} 2a \\ b \end{bmatrix}$
14	Calculate θ if the determinant of the matrix below is -16 . $\begin{bmatrix} 8 & 10 & 12 \\ 2 & \theta & 2 \\ 2 & 2 & 2 \end{bmatrix}$	14	Bereken θ indien die determinant van onderstaande matriks -16 is. $\begin{bmatrix} 8 & 10 & 12 \\ 2 & \theta & 2 \\ 2 & 2 & 2 \end{bmatrix}$

$$8(0.5b) + (-4)(3) = 2a$$

$$-2a + 4b = 12 \quad \text{--- (1)}$$

$$-4(0.5b) + 3a = b$$

$$3a = 3b$$

$$a = b \quad \text{--- (2)}$$

sub (2) in (1):

$$-2b + 4b = 12$$

$$2b = 12$$

$$b = 6 \rightarrow$$

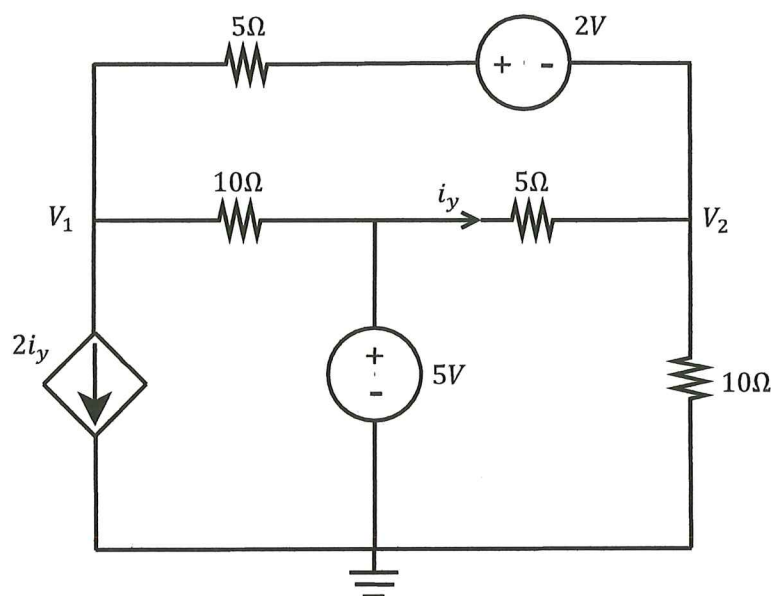
$$14) \Delta = 8[(2)(2) - (2)(2)] - 10[(2)(2) - (2)(2)] + 12[(2)(2) - (2)(\theta)]$$

$$-16 = 16\theta - 32 - 0 + 48 - 24\theta$$

$$-32 = -8\theta$$

$$\therefore \theta = 4 \rightarrow$$

Question 15-18 [4]				Vraag 15-18 [4]			
Use nodal analysis to set up two equations of the following form				Gebruik node analyse om twee vergelykings van die volgende vorm op te stel:			
$aV_1 + bV_2 = -11$				$aV_1 + bV_2 = -11$			
$cV_1 + dV_2 = 6$				$cV_1 + dV_2 = 6$			
Give the coefficients.				Gee die koëffisiënte.			
15	a	16	b	15	a	16	b
17	c	18	d	17	c	18	d



$$i_y = \frac{5 - V_2}{5}$$

$$\text{Node 1: } 2i_y + \frac{V_1 - 5}{10} + \frac{V_1 - (-2 + V_2)}{5} = 0$$

$$\times 10: 20 \left[\frac{5 - V_2}{5} \right] + V_1 - 5 + 2V_1 - 4 - 2V_2 = 0$$

$$20 - 4V_2 + 3V_1 - 2V_2 = 9$$

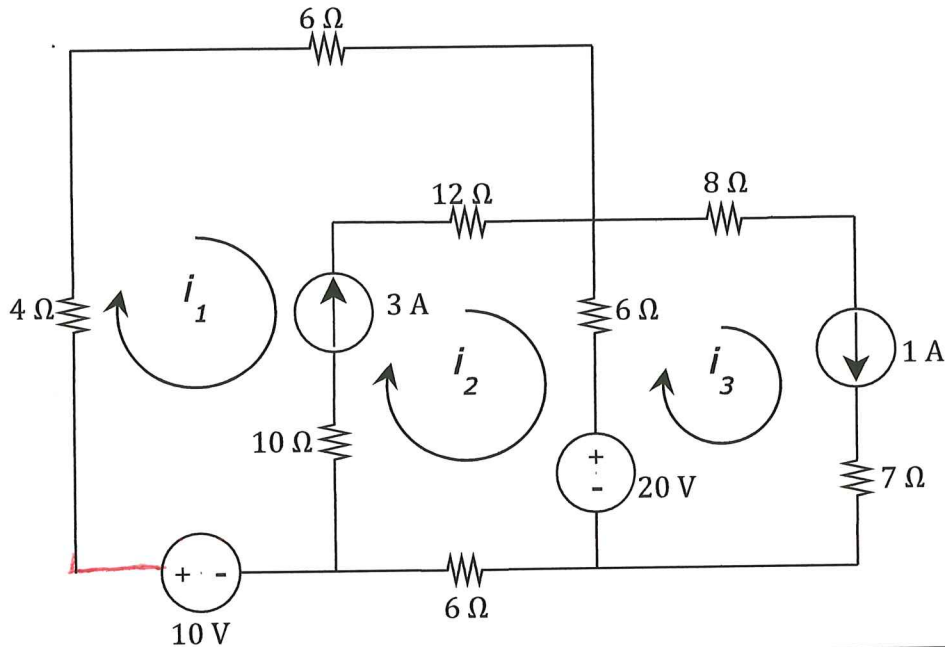
$$\underline{3V_1 - 6V_2 = -11} \rightarrow$$

$$\text{Node 2: } \frac{V_2 - 5}{5} + \frac{V_2}{10} + \frac{V_2 - (-2 + V_1)}{5} = 0$$

$$\times 10: 2V_2 - 10 + V_2 + 2V_2 + 4 - 2V_1 = 0$$

$$\underline{-2V_1 + 5V_2 = 6} \rightarrow$$

Question 19-22 [4]				Vraag 19-22 [4]			
Use mesh analysis to set up two equations of the following form:				Use mesh analysis to set up two equations of the following form:			
$ei_1 + fi_2 = -3$				$ei_1 + fi_2 = -3$			
$gi_1 + hi_2 = 4$				$gi_1 + hi_2 = 4$			
Give the coefficients.				Gee die koëffisiënte.			
19	e	20	f	19	e	20	f
21	g	22	h	21	g	22	h



$$i_2 - i_1 = 3 \Rightarrow \underline{i_1 - i_2 = -3}$$

$$SM: 10i_1 + 12i_2 - 6i_3 + 20 - 10 = 0$$

$$10i_1 + 12i_2 - 6 + 10 = 0$$

$$\underline{-10i_1 - 12i_2 = 4}$$

Question 23 [2]		Vraag 23 [2]	
23	Calculate i_a using either nodal or mesh analysis.	23	Bereken i_a deur gebruik te maak van of node- of lusanalise.

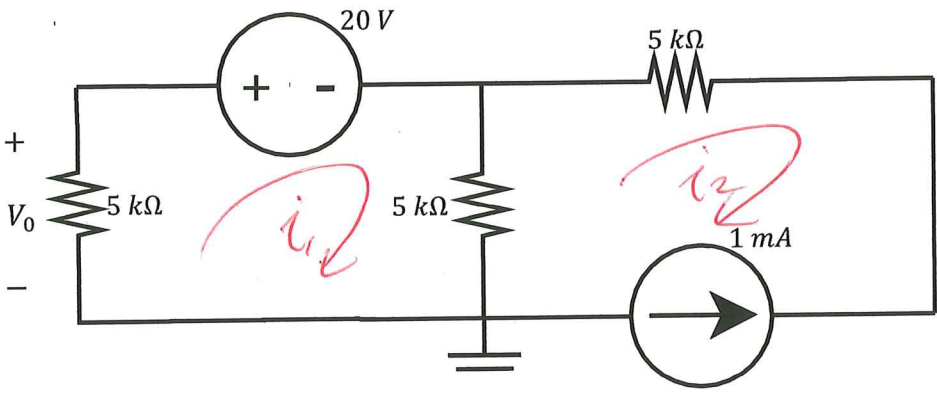
$$\text{KCL @ } V_1: \frac{V_1}{10} + \frac{V_1 - 5}{5} + \frac{2V_1}{10} + 3 = 0$$

$$\times 10: V_1 + 2V_1 - 10 + 2V_1 + 30 = 0$$

$$5V_1 = -20$$

$$V_1 = -4V$$

$$i_a = \frac{-4}{10} = -0,4 A$$

Question 24 [2]		Vraag 24 [2]	
24	Calculate V_0 using either nodal or mesh analysis.	24	Bereken V_0 deur gebruik te maak van of node- of lusanalise.
			

$$i_2 = -1 \text{ mA}$$

$$\text{KVL in } \mathcal{L}_1: 10 \text{ k} i_1 - 5 \text{ k} (-1 \text{ m}) + 20 = 0$$

$$10 \text{ k} i_1 = -25$$

$$i_1 = -2,5 \text{ mA}$$

$$V_0 = -(-2,5 \text{ m})(5 \text{ k})$$

$$= 12,5 \text{ V}$$

Question 25-28 [4]				Vraag 25-28 [4]			
Use inspection to complete the following matrix for the circuit below:				Met behulp van inspeksie, voltooi die volgende matriks vir onderstaande kringbaan:			
$\begin{bmatrix} i & -0.25 \\ -0.25 & k \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j \\ l \end{bmatrix}$				$\begin{bmatrix} i & -0.25 \\ -0.25 & k \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j \\ l \end{bmatrix}$			
25	i	26	j	25	i	26	j
27	k	28	l	27	k	28	l

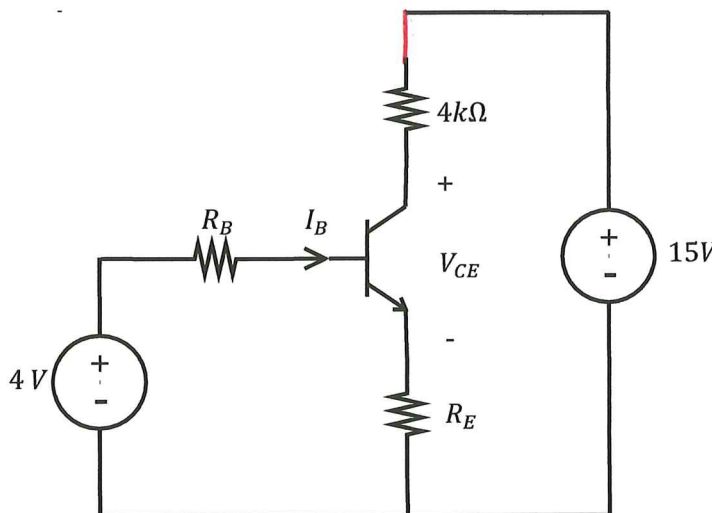
$$i = \frac{1}{5} + \frac{1}{4} = 0,45$$

$$j = 4 + 2 = 6$$

$$k = \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = 0,625$$

$$l = -2$$

Question 29 & 30 [4]		Vraag 29 & 30 [4]	
Assume $\beta = 99$ and $V_{BE} = 0.7$ V.		Aanvaar $\beta = 99$ and $V_{BE} = 0.7$ V.	
29	Determine V_{CE} if $R_E = 0 \Omega$ and $R_B = 100 k\Omega$.	29	Bepaal V_{CE} indien $R_E = 0 \Omega$ en $R_B = 100 k\Omega$.
30	Determine a value for R_B so that $V_{CE} = 5$ V, if $R_E = 1 k\Omega$.	30	Bepaal 'n waarde vir R_B sodat $V_{CE} = 5$ V, indien $R_E = 1 k\Omega$.



$$29) -V_s + I_B(100k) + 0,7 = 0$$

$$I_B = \frac{4 - 0,7}{100k} = 33 \mu A$$

$$-15 + 99(I_B)(4k) + V_{CE} = 0$$

$$\therefore V_{CE} = 1,932V$$

$$30) -15 + 99(I_B)(4k) + V_{CE} + 100I_B(1000) = 0$$

$$\text{but } V_{CE} = 5V$$

$$\therefore I_B = 20,16129 \mu A$$

$$-4 + I_B R_B + 0,7 + 100I_B(1000) = 0$$

$$\therefore R_B = 63,68 k\Omega$$

PRACTICE PAGE / OEFENBLAD 1

Assessment ID

2	0	1	7	E	B	N	1	2	2	S	1		
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Answer

Unit

01	$Saving =$	35040	R
02	$W =$	27	kJ
03	$p(0.3 s) =$	164,5	mW
04	$W =$	108,08	mJ
05	$l =$	5	No units required.
06	$n =$	7	
07	$v_I =$	15	V
08	$P_d =$	5	W
09	$R_{cd} =$	75	k Ω
10	$R_{ef} =$	33	Ω
11	$\frac{v_o}{v_s} =$	0,2	
12	$I_s =$	-0,55	A
13	$b =$	6	
14	$\theta =$	4	
15	$a =$	3	No units required.
16	$b =$	-6	
17	$c =$	-2	
18	$d =$	5	
19	$e =$	1	
20	$f =$	-1	
21	$g =$	-10	
22	$h =$	-12	

PRACTICE PAGE / OEFENBLAD 2

Assessment ID

2	0	1	7	E	B	N	1	2	2	S	1		
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Answer

Unit

23	$i_a =$	$-0,4$	A
24	$V_0 =$	$12,5$	V
25	$i =$	$0,45$	No units required.
26	$j =$	6	
27	$k =$	$0,625$	
28	$l =$	-2	
29	$V_{CE} =$	$1,932$	V
30	$R_B =$	$63,68$	k- Ω